Team Name: Team SolarCycle

Robust, Solar-Powered Charging Infrastructure for a Shared e-Bike Scheme under Demand Uncertainty

Theme 3 (primary): solar-PV charging-station design · Theme 2 (secondary): managed charge/discharge profiles | Example site: University of Warwick, Coventry (CV4 7AL)

Approach (299 words)

Aim. We design a solar-powered charging hub for a shared e-bike scheme at the University of Warwick, Coventry (CV4 7AL). The hub must stay affordable and reliable across every realistic future demand, not just an average forecast (Theme 3). We also model how the batteries charge and discharge on different routes and compare private versus shared e-bikes (Theme 2).

Objectives. (1) build realistic demand scenarios for the scheme; (2) size the solar PV, battery and charge bays behind a limited grid connection; (3) measure the cost and service gained by designing for robustness; (4) test whether a battery is worth its cost.

Research questions. How large should the hub be when demand is uncertain? What does robustness save against a normal average-demand design? Does a battery pay at a grid-connected hub? How do charge and discharge profiles differ between private and shared e-bikes?

Methodology (Figure 1). We build Low, Medium and High demand scenarios from real DfT shared-micromobility monitoring data (all UK trial areas), adapted to a UoW e-bike scheme, and scale them across a growth range into nine weighted scenarios; the official UoW Bikes file is used when provided. A physics-based route model (Burani 2022; Ouf 2023) gives the energy each ride draws from gradient, speed and assist, and produces the Theme 2 charge/discharge profiles. An hourly model (8,760 h) sends solar to demand, then to the battery, then to the capped grid. We score all 150 designs (PV 5-25 kWp, battery 0-50 kWh, 4-20 bays) under five decision rules (naive, two-stage stochastic, CVaR, minimax-regret, maximin) and map the cost-versus-robustness frontier. We then test the result with a 500-run Monte-Carlo, Sobol sensitivity, a penalty-grid-horizon robustness sweep, and validation against published data. Solar (PVGIS, Coventry) and grid carbon (National Grid ESO, West Midlands) are real API data.

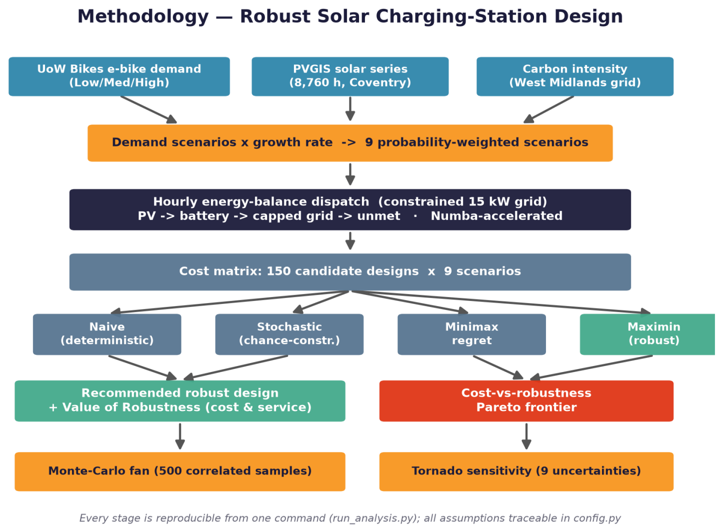

Figure 1 — End-to-end methodology / algorithm flow.

Outcomes (455 words)

Recommended design. The model recommends 15 kWp of solar, no battery, and 8 smart-managed charge bays, costing about £29,400 (Figure 2).

Value of robustness. A normal design sized to average demand (5 kWp, 4 bays) looks cheap but fails when demand is high: its worst-case annual cost is £47,179 and it serves only 95.3% of charging. The robust design holds worst-case cost to £19,232/yr, a 59% (£27,947) cut, and keeps 99.6% service in every scenario. This is not a strawman: the same search also produced stochastic and CVaR designs, and the robust (maximin) design still gives the lowest worst-case cost. Across the Monte-Carlo fan its 95th-percentile cost is £10,903 against £10,126, roughly halving the downside. It beats the naive design in 100% of 35 penalty-grid cases (9/9 benchmarks validated).

Does a battery pay? Because charging is flexible, smart scheduling flattens the load below the grid limit, so a battery never enters the robust design at the 15 kW connection. A dedicated grid sweep shows a battery only pays at about 9 kW or weaker (toward off-grid). The greenest and cheapest answer is solar plus smart bays, which also avoids the embodied carbon of an unneeded battery.

Theme 2: routes and profiles (Figure 3). The route model gives 4-18 Wh/km depending on gradient, speed and load (campus hops lowest, hilly and cargo trips highest). The profiles differ clearly: a private e-bike makes two longer trips and takes one deep overnight home charge (127 full cycles/yr), while a shared bike runs many short trips, discharges deeper and is topped up at the depot (169 cycles/yr). Shared batteries work about 1.3x harder and age faster, which is why a shared scheme needs the managed hub and long-life LFP cells.

Sustainability. The hub avoids 3.7 tCO_2/yr, about 15% of grid-only emissions on a consequential (marginal, gas-margin) basis, with 15% of demand met directly by solar. The LFP storage option (weak-grid case) is cobalt and nickel free, with about 45% of pack mass recoverable.

Output format (GUI). An interactive dashboard lets a planner set PV, battery, bays and the grid limit with sliders and see the hourly energy balance, service level, annual cost and carbon update live, alongside the Pareto frontier and Monte-Carlo fan. One command also writes results.json, every figure and this summary.

Originality & reproducibility. All code was written for this project. Every number is reproducible from one command with a fixed seed and an automated test suite. Solar and grid carbon are real API data; demand is built from real DfT shared-micromobility monitoring data (Open Government Licence), validated against published ranges, and replaced by the official UoW Bikes file when supplied.

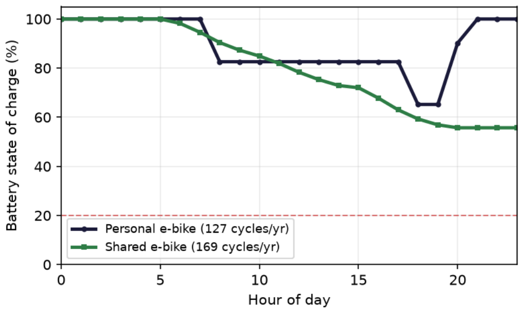

Figure 2 — Cost-vs-robustness Pareto frontier (Theme 3). Figure 3 — Personal vs shared e-bike charge/discharge profiles (Theme 2).

Links to GitHub files

GitHub (executable code + README / QuickStart guide): https://github.com/Gourav88502/emicromobility-robust-charging — the README documents the one-command run ("python run_analysis.py"), data sources and how to drop in the official UoW Bikes Data(Sheet1).csv.

References (not counted in the word limit)

[1] Burani, E., Cabri, G., Leoncini, M. (2022). An Algorithm to Predict E-Bike Power Consumption Based on Planned Routes. Electronics, 11(7), 1105.

[2] Ouf, K., Soubra, H., Mazhr, A. (2023). E-Bike Energy Needs Estimation based on Route Characteristics and Rider Behavior. IEEE ICICIS 2023, 345–352.

[3] Corti, F. et al. (2024). A comprehensive review of charging infrastructure for Electric Micromobility Vehicles. Energy Reports, 12, 545–567.

[4] Marie, J.-J. (2023). The Micromobility Revolution Gathers Momentum. Faraday Insights, Issue 16.

[5] Gössling, S. (2020). Integrating e-scooters in urban transportation. Transportation Research Part D, 79, 102230.

[6] Rockafellar, R.T., Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. Journal of Risk, 2(3), 21–41.

[7] Birge, J.R., Louveaux, F. (2011). Introduction to Stochastic Programming, 2nd ed. Springer.

[8] PVGIS (EU JRC) hourly series, University of Warwick (52.3838, −1.5616); National Grid ESO Carbon Intensity API, West Midlands (region 8).

[9] Cost & performance benchmarks: BEIS, IRENA, BloombergNEF, IEA PVPS, Fraunhofer ISE — see REFERENCES.md.